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# Vector calculus <sup>mod.</sup> exam.

$$1. (a) \nabla \times B = \begin{pmatrix} \partial_y B_z - \partial_z B_y \\ \partial_z B_x - \partial_x B_z \\ \partial_x B_y - \partial_y B_x \end{pmatrix} = \begin{pmatrix} ay - 0 \\ 0 - 0 \\ \cos y - cx \cos y \end{pmatrix}$$

$$\begin{cases} ay = ay \\ bx^2 = 0 \\ (1-x) \cos y = (1-cx) \cos y \end{cases} \Rightarrow \begin{cases} a=1 \\ b=0 \\ c=1 \end{cases}$$

$$\nabla \cdot A = 0$$

(b) any of a & b

$$(c) (i) \nabla \times F = \begin{pmatrix} 0 - 0 \\ 2z \sinh x + \beta z \sinh x \\ -2 \sinh x - \alpha \sinh x \end{pmatrix}$$

$$\begin{cases} \beta + 2 = 0 \\ \alpha + 2 = 0 \end{cases} \Rightarrow \alpha = \beta = -2$$

$$(ii) \nabla \phi = (- (ry - z^2) \sinh x, r \cosh x, -2z \cosh x)$$

$$\begin{cases} r = -\alpha \\ r = 2 \\ \beta = -2 \end{cases} \Rightarrow \begin{cases} \alpha = -2 \\ \beta = -2 \\ r = 2 \end{cases}$$

$$(iii) \int_c F dr = F(r(1)) - F(r(0)) = -4$$

$$2. \quad (a) \quad F = \nabla f = (2x, 2y, 1)$$

$$\nabla \times F = \begin{pmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ x & y & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \vec{0}$$

$$\nabla \cdot F = 4$$

$$(b) \quad f = \rho^2 + z \quad F = (2\rho \cos \phi, 2\rho \sin \phi, 1)$$

$$(c) \quad \nabla f = (2\rho, 0, 1)$$

$$\nabla^2 f = 4$$

$$3. \quad (a) \quad \int_S \nabla \times A \cdot \vec{n} \, da = \int_C A \cdot d\vec{l}$$

$$(b) \quad \int_C \vec{r} \cdot d\vec{r} = \int \nabla \times \vec{r} \cdot \vec{n} \, da = 0$$

$$\begin{aligned} (c) \quad I &= \oint_C (y, -x, 0) \, d\vec{r} \\ &= \int_0^{2\pi} (\sin t, -\cos t, 0) \cdot (-\sin t, \cos t, 0) \, dt \\ &= -2\pi \end{aligned}$$

$$\begin{aligned} (d) \quad I_z &= \iint_S \vec{G} \cdot \vec{n} \, dS = \iint_S (x, y, -2-2z) \cdot (x, y, z) \, dS \\ &= \iint_S x^2 + y^2 - 2z - 2z^2 \end{aligned}$$

$$= \int_0^{\frac{\pi}{2}} \int_0^{\pi} (1 - \cos\theta - 3\cos^3\theta) \sin\theta \, d\theta \, d\phi$$

$$= -2\pi$$

(e)  $G = \nabla \times A$

Electromagnetism mode paper-

4. (a)  $\nabla \times E = 0$

$$E = -\nabla V$$

$$V = - \int_0^1 E(r(t)) \frac{dr}{dt} \cdot dt$$

$$= \alpha e^{-\rho r^2}$$

1b) (ii)  $\iint_{\partial V} E \cdot d\vec{a} = \frac{q_{\text{enc}}}{\epsilon_0} = \frac{2L\lambda}{\epsilon}$

(iii)  $E \cdot 2\pi R \lambda L = \frac{\lambda L}{\epsilon}$

$$E = \frac{\lambda}{2\pi R \epsilon}$$

5. (a)  $\frac{\rho}{\epsilon} = \nabla \cdot E = 0$

1b)  $\nabla \times E = -\frac{\partial}{\partial t} B$

$$\nabla \times E = \dots$$

Mode exam.

2. a)  $F = -\nabla f$

$$-\nabla f = (-2bxz, -2ayz, -bx^2 - ay^2 - 2cz)$$

$$\begin{cases} -2bxz = -2xz \\ -2ayz = 2yz \\ y^2 - x^2 + 4z = -bx^2 - ay^2 - 2cz \end{cases} \Rightarrow \begin{cases} a = -1 \\ b = 1 \\ c = -2 \end{cases}$$

1b)  $\int_c \vec{F} \cdot d\vec{r} = \int_c -\nabla f \cdot d\vec{r}$

$$r(2\pi) = (4, 0, 1) \quad = -\left(f[r(2\pi)] - f[r(0)]\right)$$

$$r(0) = (1, 0, 0) \quad = -(-1) = 1$$

$$f = x^2z - y^2z - 2z^2$$

1c)  $\int_c \vec{F} \cdot d\vec{r} = 0$  since  $r(0) = r(2\pi)$

2. a)  $\nabla f = (1 - yz \sinh(xy), -2y - xz \sinh(xy), \cos(xy))$

$$\nabla^2 f = -y^2z \cos(xy) - 2 - x^2z \cos(xy) + 0$$

$$= -(x^2 + y^2)z \cos(xy) - 2$$

$$\begin{aligned}
 1b) \quad \nabla \cdot g^h A &= \partial_i g^h A_i \\
 &= A_i \partial_i g^h + g^h \partial_i A_i \\
 &= A_i (h \partial_i g + g \partial_i h) + g^h \partial_i A_i \\
 &= g^h \nabla \cdot A + h A \nabla g + g A \cdot \nabla h
 \end{aligned}$$

$$\begin{aligned}
 \nabla \times (g A) &= \epsilon_{ijk} e_i \partial_j (g A)_k \\
 &= \epsilon_{ijk} e_i (A_k \partial_j g + g \partial_j A_k) \\
 &= \nabla g \times A + g \nabla \times A
 \end{aligned}$$

$$1c) \quad \vec{r} = (u-v, u+v, w)$$

$$h_u = \left| \frac{\partial \vec{r}}{\partial u} \right| = |(1, 1, 0)| = \sqrt{2}$$

$$h_v = \left| \frac{\partial \vec{r}}{\partial v} \right| = |(-1, 1, 0)| = \sqrt{2}$$

$$h_w = 1$$

$$dV = h_u h_v h_w \, du \, dv \, dw = du \, dv \, dw$$

$$\begin{aligned}
 3. \quad 1a) \quad (i) \quad \int_V \nabla \cdot \vec{F} \, dV &= \int_{\partial V} \vec{F} \cdot \vec{n} \, da \\
 &= \int_{side} (x, y, z) \cdot \left( \frac{x}{a}, \frac{y}{a}, 0 \right) da \\
 &\quad + \int_{up-top} (x, y, z) \cdot (0, 0, 1) da \\
 &\quad + \int_{low-top} (x, y, z) \cdot (0, 0, -1) da
 \end{aligned}$$

$$\begin{aligned}
&= \int_{\text{side}} \frac{1}{a} (2x^2 + y^2) da + 0 \\
&= \frac{1}{2a} \int_{\text{side}} (3x^2 + 3y^2) da \\
&= \frac{3a^2}{2a} \cdot \pi a \cdot 2h = 6\pi h a^2
\end{aligned}$$

(b) Stokes' theorem.

$$\iint_S \nabla \times \mathbf{F} \cdot \vec{n} da = \int_{\partial S} \mathbf{F} \cdot d\mathbf{l}$$

$$\begin{aligned}
\text{LHS} &= \iint_S \begin{pmatrix} \partial_y A_z - \partial_z A_y \\ \partial_z A_x - \partial_x A_z \\ \partial_x A_y - \partial_y A_x \end{pmatrix} \cdot (x, y, z) da \\
&= \iint_S (-yz + zy, 0 - 0, 0 + 1) \cdot (x, y, z) da \\
&= \iint_S z da = 0
\end{aligned}$$

$$\begin{aligned}
\text{RHS} &= \int_{\partial S} \mathbf{F} \cdot d\mathbf{l} \quad \text{where } \partial S = \phi \\
&= 0
\end{aligned}$$

$$4 \quad (a) \quad \nabla \times \mathbf{E}_1 = \alpha (0-1, 0-1, 0-1) = \alpha (-1, -1, -1) \neq 0$$

$$\nabla \times \mathbf{E}_2 = \alpha (0-0, 0-0, 1-1) = \vec{0}$$

$$(b) \quad \mathbf{E} = -\nabla V$$

$$V = \int_{r_0}^r \nabla V \cdot d\mathbf{l} = \int_{(0,0,0)}^{(x,y,z)} -\mathbf{E} \cdot d\mathbf{l}$$

$$\begin{aligned}
&= \int_0^1 -(x+y, x+y, 0) \cdot (x, y, z) dt \\
&= - \int_0^1 x^2 + 2xy + y^2 dt \\
&= -(x+y)^2
\end{aligned}$$

$$\begin{aligned}
1c) \quad \nabla \cdot E &= \frac{\rho}{\epsilon_0} \\
&= 2\alpha
\end{aligned}$$

$$\rho = 2\alpha \epsilon_0$$

$$\begin{aligned}
1d) \quad \int_V E \cdot d\vec{a} &= \int_V \nabla \cdot E \, dV \\
&= 2\alpha \epsilon_0 \cdot 8b^3 = 16\alpha \epsilon_0 b^3
\end{aligned}$$

$$5. 1a) \quad q_f = \frac{4}{3} \pi R^3 \rho_f$$

$$1b) \quad \nabla \cdot D = \rho_f$$

$$\begin{aligned}
\int_{\partial B_r(0)} D \cdot d\vec{a} &= \int_{\partial B_r(0)} D(r) \cdot \hat{r} \, da = \int_{B_r(0)} \nabla \cdot D \, dV \\
&= 4\pi r^2 D(r) = \frac{4}{3} \pi R^3 \rho_f
\end{aligned}$$

$$D(r) = \frac{R^3 \rho_f}{3r^2 \epsilon}$$

$$1c) \quad D = \epsilon_0 (1 + \chi_e) E$$

$$\rho = \epsilon_0 \chi_e E = \epsilon_0 \chi_e \frac{D}{\epsilon_0 (1 + \chi_e)} = \frac{\chi_e}{1 + \chi_e} D$$



$$\begin{aligned}
 p_b &= -\nabla \cdot \vec{p} = -\frac{\chi_e}{1+\chi_e} \nabla \cdot \left( \frac{R_1^2 \mu_f}{3r^3} \hat{r} \right) \\
 &= -\frac{\chi_e}{1+\chi_e} \frac{R_1^2 \mu_f}{3} \left( \frac{3}{r^3} + r \frac{d}{dr} \frac{1}{r^3} \right) \\
 &= 0
 \end{aligned}$$

$$\vec{\sigma}_b = p \cdot \vec{n}$$

$$\begin{aligned}
 6. (a) \quad \nabla \cdot \vec{B}_1 &= 0 \\
 \nabla \cdot \vec{B}_2 &= \alpha f \neq 0
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad \nabla \times \vec{B} &= \mu_0 \vec{J} \\
 &= \begin{pmatrix} 0 & -0 \\ 0 & -0 \\ 1 & -(-1) \end{pmatrix} = (0, 0, 2) \\
 \vec{J} &= 2\mu_0 \hat{z}
 \end{aligned}$$

$$(c) \quad \vec{B} =$$